

Arm Education: Internet-of-Things Education Kit

Lectures

General note: Update copyright footer on all slides

Module 1: Introduction to the Internet of Things

- Slide 13:
 - Remove reference to “Mbed”, and replace with “CMSIS”
- Slide 14:
 - Remove Arm Mbed graphic (bottom right), and replace with CMSIS branding
- Slide 24:
 - Remove Arm Mbed graphic (bottom right), and replace with CMSIS branding, as from Slide 14
- Slide 25:
 - Replace “Arm Mbed: Accelerating IoT deployment” with “CMSIS: Accelerating IoT Deployment” in slide subtitle.
 - Add “, and entry-level Cortex-A application processors” after “Cortex-M microcontrollers”
 - Delete the third bullet-point and its sub-bullet-points completely (“Simplify integration with...”)
- Slide 30:
 - Replace key points in text (bottom right):
 - IoT has been around for about 40 years
 - ... number of connected things (22bn in 2026)
 - ... world’s human population (8.3bn in 2026)

Module 2: IoT System Architecture and Standards

- Slide 3:
 - Insert “own” after “its” and before “unique” on the second bullet-point: “... has its own unique requirements ...”
- Slide 5:
 - Replace “... enables to break up complexity ...” with “... enables breaking up complexity ...”
- Slide 6:
 - Change “layer” to “layered” in the slide title, so it reads: “Advantages of the layered approach”
- Slide 7:
 - Remove “s” from “ecosystems” in slide subtitle.
 - Add “and virtual machines” after “containers” in bullet-point 5
- Slide 8:
 - Change “ZigBee” to “Zigbee” (the B is not capitalised)
- Slide 11:
 - Remove “for” before “e.g.” in the first box.
- Slide 12:
 - Remove “for” before “e.g.” in the first box
 - Remove “for” before “e.g.” in the fourth box
- Slide 16
 - Change “... down at device level” to “... down to the device level”
- Slide 19
 - Change “ZigBee” to “Zigbee” (the B is not capitalised)
- Slide 22:
 - Change “ZigBee” to “Zigbee” (the B is not capitalised)

Module 3: Introduction to Embedded Systems

- Slide 3:
 - Remove “for” before “e.g.”
- Slide 7:
 - Change “Outputs” to:
 - Display (current speed, heart rate, etc)
- Slide 8 for me contains some odd bullet-point indentation, but it could be the presentation viewer that I’m using (I don’t have PowerPoint) – but it’s worth a quick check that alignment is consistent.
- Slide 13 similarly has a text alignment issue for me – the second column text starts lower than the first column, so the layout looks a bit odd.
- Slide 15:
 - Change “... rather than in Java that results in ...” to “... rather than in Java, which results in ...”

Module 4: Hardware Platforms for IoT

- Slide 11:
 - There’s an odd bullet-point marking in the middle of the text – it looks like the main text is an imported image, rather than proper text. This may need to be corrected, or at least the spurious bullet-point removed.
- Slide 15:
 - Delete “move” from before “wake-up” from the second box.
- Slide 16:
 - Similar issue to slide 11 – spurious bullet-point, and image instead of real text.
- Slide 20:
 - Also a spurious bullet-point

Module 5: The Arm Cortex-M4 Processor Architecture

- Slide 3:
 - The fourth point states “Arm does not manufacture,...” – but a recent news bulletin indicates that Arm is starting to manufacture its own semiconductors, so this point may need to be amended.
 - Please check the details of this, but if it is the case, then the point could at least become:
 - “Arm historically has not manufactured its own processes, instead licensing designs to semiconductor companies, who customize their own Intellectual Property (IP) on top of Arm’s.”
 - If you feel it’s appropriate to be explicit, then the point could be:
 - “Arm has historically only licensed processor IP to other companies (who customise with their own IP), but are now fabricating their own semiconductors”
- Slide 5:
 - Add “brace” underneath Armv4 to Armv7 to indicate these are 32-bit architectures
 - Add “Armv9” after “Armv8”
 - Add “brace” underneath Armv8 to Armv9 to indicate these are 64-bit architectures
 - **See provided reference slide for more info**
- Slide 20:
 - Add “, and holding the values of local variables in functions” to “R13: SP” bullet-point #2, after “switching between tasks”
- Slide 21:
 - Change “PC” to “R15: PC - Program Counter” in main bullet-point
- Slide 22:
 - Change “R14: LR” to “R14: LR - Link Register” in main bullet-point
- Slide 25:
 - Replace “PRIASK” with “PRIMASK” (missing M) in first sub-bullet-point

Module 6: Interrupts and Low Power Features

- Slide 3:
 - Change “Logic Solution” to “Logical Solution” in 3rd bullet-point
- Slide 5:
 - Expand “ISR” to “Interrupt Service Routine (ISR)” in second sub-bullet-point
 - Change “Interrupt routine” to “Interrupt Service Routine” in figure, above right-most box.
- Slide 8:
 - Change “instructions” to “instruction” (remove ‘s’) in first bullet-point
- Slide 41:
 - Remove references to Mbed Platform, and replace with CMSIS:
 - Introduction to CMSIS and Keil Extensions for VS Code

Module 7: Introduction to the Mbed Platform

- Replace module with reference slides (will need formatting to be consistent with existing style)

Module 8: IoT Connectivity Part 1

- Slide 2:
 - Replace “ZigBee” with “Zigbee”
- Slide 4:
 - Replace “Master <-> Slave architecture” with “Central <-> Peripheral architecture”
 - Replace “Master can have up to seven active slave devices” with “Central controller can have up to seven active peripheral devices”
 - Replace “... set-up by a master ...” with “... set-up by a central controller ...”
 - Replace “... act as slaves in ...” with “... act as peripherals in ...”
 - Replace “... and as master in ...” with “... and as a central controller in ...”
 - The terms used in the graphic need replacing:
 - Master -> Central Controller
 - Slave -> Peripheral
- Slide 5:
 - Replace all instances of “Master” with “Central controller”
 - Replace all instances of “Slave” with “peripheral”
- Slide 6:
 - Replace “... establishment between master/slave” with “... establishment between central/peripheral”
- Slide 8:
 - Replace “ZigBee” with “Zigbee” (second bullet-point)
- Slide 9:
 - Replace “... part of the master’s address ...” with “... part of the central controller’s address ...”
- Slide 12:
 - Replace “... of the master” with “of the central controller”
 - There is an inconsistent bullet-point spacing in the “Sync words” column on the “XOR with...” line
- Slide 13:
 - Replace “... seven active slaves ...” with “... seven active peripherals ...”
- Slide 14:
 - Similar to other modules, this slide appears to be an imported image, and should be replaced with proper text. There is an out-of-place bullet-point mark due to this.
 - Once converted to text, replace:
 - “... the master’s UAP” with “... the central controller’s UAP”

- “... the master’s clock” with “... the central controller’s clock” in the second and fourth bullet-points
- Slide 17:
 - In the first bullet-point:
 - Replace “Master aiming ...” with “Central controller aiming ...”
 - Replace “(slaves)” with “(peripherals)”
 - The state diagram will need the text adapted to change references:
 - Master -> Central
 - Slave -> Peripheral
- Slide 18:
 - Replace “... from the slave device ...” with “... from the peripheral device ...” in the third bullet-point
 - Replace “Master that ...” with “Central controller that ...” in the fifth bullet-point
- Slide 19:
 - Replace “Master hops on the sequence derived from the slave’s address” with “Central hops on the sequence derived from the peripheral’s address”
 - Replace “... from the slave’s address” with “... from the peripheral’s address” (second bullet-point)
 - Replace “Slave responds ...” with “Peripheral responds ...” (third bullet-point)
 - Replace “Master sends ...” with “Central controller sends ...” and “... the master’s address ...” with “... the central controller’s address ...” (fourth bullet-point)
 - Replace “Master assigns ...” with “Central controller assigns ...” (fifth bullet-point)
 - The graphic on the right needs text adapted to change references:
 - Master -> Central
 - Slave -> Peripheral
- Slide 20:
 - Replace “Number of slaves can be attached to master ...” with “Number of peripherals that can be attached to master ...”
- Slide 21:
 - Completely remove first bullet-point:
 - “The master of a connection is referred to as “central” and the slave as “peripheral”
- Slide 27:
 - Remove “(master)” from first bullet-point
 - Remove “(slave)” from second bullet-point
 - Replace “Master” with “Central” and “Slave” with “Peripheral” in the third bullet-point
- Slide 35:
 - This slide comprises an image, and as such suffers the spurious bullet-point. It will need to be converted to text for accessibility.
- Slide 37:
 - Replace “ZigBee” with “Zigbee” in slide title, and in device type bullet-points.
- Slide 38:
 - The graphic will need corrected to re-style “ZigBee” to the correct branding: “Zigbee”
- Slide 42:
 - Replace “ZigBee” with “Zigbee” in third bullet-point

Module 9: IoT Connectivity Part 2

- Slide 21:
 - This image will need to be converted to text
- Slide 22:
 - There is a spurious bullet-point in the second column, where an image is used.

Module 10: The Cloud

- Slide 5:
 - Change “Machines” to “Machine” in slide title
- Slide 29:
 - Remove “Mbed Linux OS” from left-most list
 - Replace “Mbed OS” with “CMSIS-RTOS” in left-most list

Module 11: IoT Security

- Slide 19:
 - Remove “for” in “for e.g.” (first bullet-point)
- Slide 21:
 - Remove “for” in “for e.g.” (second box)
- Slide 22:
 - Remove “for” in “for e.g.” (fourth bullet-point)
 - Remove “(hash)” at the end of the fifth bullet point
- Slides 32, 33, 34, 35, 36, 37
 - These slides contain an image in place of the content that will need to be converted to text for accessibility. As in other slides, this causes a spurious bullet-point to appear at the top.
- Slide 39
 - Change “Pairwise Master Key” to “Pairwise Main Key”

Module 12: Current and Future Trends of IoT

- Slide 3:
 - Update infographic on right to:
 - 21bn Internet-connected devices at the end of 2025
- Slide 6:
 - Change “... the brain perception process in a brain ...” to “... the perception process in a brain ...”
 - (Remove superfluous “brain”)
- Slide 7:
 - This slide needs to be converted to text, and the spurious bullet point removed.
- Slide 9:
 - Change “TensorFlow Light” to “TensorFlow Lite”
- Slide 12:
 - The graphic should be drawn natively, as it is currently a low-quality image